

# STACK MCQ XML Generator

## macOS Setup Guide

From Homebrew preparation and Git clone to opening localhost:4173

**Audience** For users starting from a fresh Git clone and opening the local web app for the first time.

### What you will accomplish

- Install or verify Homebrew, Git, Python, and Maxima.
- Clone stack\_questions from GitHub.
- Configure STACK code and enable local Maxima evaluation.
- Open the local web app in your browser.

**Note** An internet connection is required for the initial setup. No additional Python packages are required.

### Quick path

|   |                                                                    |
|---|--------------------------------------------------------------------|
| 1 | Open Terminal and prepare Homebrew.                                |
| 2 | Run brew install python maxima.                                    |
| 3 | Clone stack_questions.                                             |
| 4 | Run the macOS launcher.                                            |
| 5 | Press Enter to download STACK automatically.                       |
| 6 | Open <a href="http://localhost:4173/">http://localhost:4173/</a> . |

# 1. Prepare Homebrew, Python, and Maxima

## 1 Open Terminal

In Finder, open **Applications - Utilities - Terminal**. Paste the following commands one line at a time.

## 2 Check Command Line Tools

Run the command below. If the tools are not installed, macOS opens Apple's installer. Git becomes available after installation.

```
xcode-select --install
```

## 3 Install Homebrew

If the brew command is unavailable, use the installation command from the official Homebrew website. Review the prompt and enter your Mac login password when requested.

<https://brew.sh/>

```
/bin/bash -c "$(curl -fsSL
https://raw.githubusercontent.com/Homebrew/install/HEAD/install.sh)"
```

**Important** If Homebrew displays Next steps, run the shellenv command shown there and then reopen Terminal.

## 4 Install Python and Maxima

Update Homebrew, then install Python 3 and Maxima.

```
brew update
brew install python maxima
```

## 5 Verify installation

Each command should print a version number:

```
git --version
python3 --version
maxima --version
```

Official information: <https://git-scm.com/install/mac> / <https://www.python.org/downloads/macos/> / <https://maxima.sourceforge.io/download.html>

## 2. Clone and perform first-time setup

### 1 Clone stack\_questions

The following example stores the repository in your Documents folder:

```
cd "$HOME/Documents"
git clone https://github.com/yositomi-opu/stack_questions.git
cd stack_questions
```

### 2 Run the macOS launcher

From the repository root, run the command below. You can also double-click **scripts - macos - start-mcq-webapp.command** in Finder.

```
./scripts/macos/start-mcq-webapp.command
```

### 3 Choose the STACK location

At the first **STACK location** prompt, press Enter without typing anything if you do not already have STACK. The launcher downloads moodle-qtype\_stack automatically.

```
STACK code setup is required.
If STACK is not cloned yet, press Enter to download it automatically.
STACK location: [press Enter here]
```

#### Existing STACK clone

If you already have moodle-qtype\_stack, enter its full folder path, for example  
/Users/yourname/work/moodle-qtype\_stack.

### 4 Wait for the automatic check

Setup is complete when **STACK code: OK** and the local CAS availability message appear. The browser then opens automatically.

```
Python: OK
Maxima: OK (...)
STACK code: OK
STACK loading mode: dumped executable
The MCQ WebApp local CAS evaluation is available.
```

#### Normal fallback

If dump generation is unavailable, the launcher automatically switches to loading STACK at evaluation time. This is expected behavior.

### 3. Open the web app

#### 1 Confirm browser startup

After setup, your default browser opens. If it does not, enter either URL in Safari or Chrome:

```
http://localhost:4173/
http://127.0.0.1:4173/
```

#### 2 Verify STACK evaluation

In Problem variables, enter an expression containing `rand([1, 2, 3])`, then click **Evaluate problem variables**. If 1, 2, or 3 appears, STACK is working.

#### 3 Stop the server

Select the Terminal window used by the launcher and press **Control+C**.

## Troubleshooting

| Symptom                                  | What to do                                                                                    |
|------------------------------------------|-----------------------------------------------------------------------------------------------|
| permission denied                        | Run <code>chmod +x scripts/macos/start-mcq-webapp.command</code> .                            |
| Developer cannot be verified             | Control-click the launcher in Finder and choose Open.                                         |
| brew: command not found                  | Complete Homebrew's Next steps and reopen Terminal.                                           |
| Maxima is not found                      | Run <code>brew install maxima</code> and verify <code>maxima --version</code> .               |
| Address already in use                   | The server is already running. Open <code>localhost:4173</code> .                             |
| STACK code: not loaded                   | Run the launcher again and provide the STACK location, or press Enter for automatic download. |
| <code>rand(...)</code> remains unchanged | Run <code>--check</code> , restart with <code>--reload</code> , and reload the browser page.  |

**More help** See `app/mcq-webapp/README.md`. Run `python3 app/mcq-webapp/server.py --check` for diagnostics, or add `--reload` to restart.